

Colony PCR

Overview

Rapidly screen individual bacterial colonies to verify the presence of the desired DNA insert or recombinant construct using the colony as the PCR template.

A. Materials

- a. Plate with bacterial colonies
- b. Empty plate (depends on antibiotics selection)
- c. Individual PCR tube or PCR 8-strip tubes
- d. Forward primer (10 μ M)
- e. Reverse primer (10 μ M)
- f. Q-Amp 2X ScreeningFire Taq Master Mix
- g. ddH₂O
- h. PCR machine

B. Procedure

1. Label the empty plate and draw a grid on it (based on how many colonies).
2. Prepare the following reaction(s) based on the amount of colonies.

	1 Reaction	Final Volume
Q-Amp 2X ScreeningFire Taq Master Mix	1X	10 μ L
Forward primer (10 μ M)	0.5 μ M	1 μ L
Reverse primer (10 μ M)	0.5 μ M	1 μ L
ddH ₂ O (Top up to total volume of 20 μ L)	-	8 μ L
Total Volume	-	20 μ L

3. Pipette 20 μ L of the master mix to each PCR tube.
4. Pick up a colony with a micropipette tip.
5. Streak the bacteria onto the corresponding grid box .
6. Dispense the tip in a PCR tube.
7. Repeat this step for all colonies.
8. Place into PCR machine for thermal cycle.

C. Run PCR program

Stages	Temperature (°C)	Time (sec)	Cycles
Initial Denaturation	95°C	2 min	1 cycle
Denaturation	95°C	30 sec	30 cycles
Annealing	65°C	40 sec	
Extension	72°C	2.5 min	
Final extension	72°C	1 min	1 cycle

Colony PCR

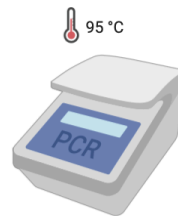
- 1 Pick colonies without fluorescence with a sterile pipette



- 2 Add colony to PCR tubes with master mix



- 3 Run PCR thermal cycle



- 4 Run DNA gel, check for expected colony PCR products

